RPMGLOBAL

AMT - Dashboards (KPIs and Accountability)

Training Manual

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RPM SOFTWARE - ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. The guide is constructed in a way that teaches students the key functionality of the application through practical hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

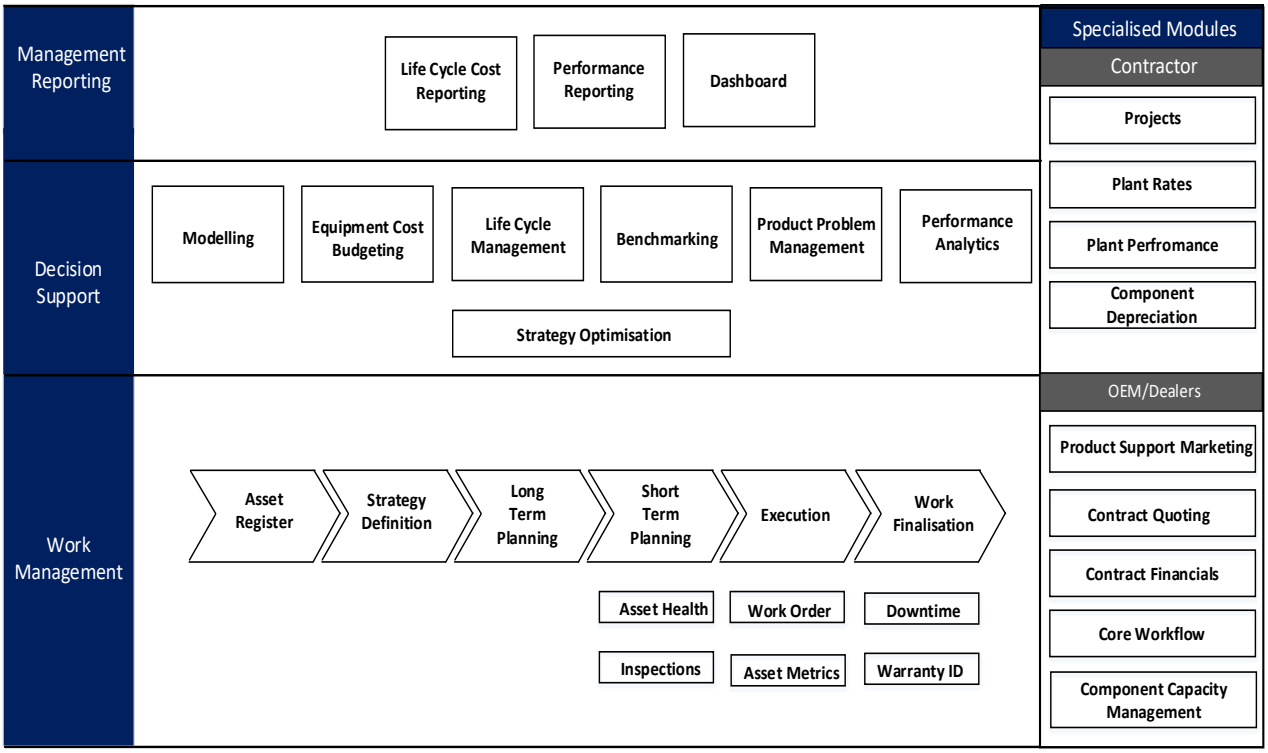
Check your understanding on the workshop and ask any related questions.

Course Outline

Pre-requisites	
Target Audience	Senior Management, Financial Controller, Maintenance Manager, Strategist, Reliability Engineer, Long Term Planner, Short Term Planner, Fleet Analyst, Administrator, Contract Manager, Contract Accountant, Sales Representative, Sales Manager, Marketing Director.
Nominal Duration	
Course Objectives	
Learning Outcomes	
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

Introduction to AMT

AMT is a comprehensive asset management system consisting of integrated modules as depicted below. The functionality covered by this training manual is identified by the **Dashboard** section in the diagram.



OVERVIEW OF DASHBOARDS

The Dashboard is a management system that enables organisations to clarify their vision and strategy and translate them into action. It provides feedback on both the internal business processes and external outcomes in order to improve strategic performance and results. When fully deployed, the Dashboard places strategic planning at the centre of an enterprise. By establishing measurements covering multiple aspects of AMT operations, performance can be continuously monitored. The use of progressive alarm signals enables exception reporting to be clearly focussed.

Training Workshops

1. System Configuration and Settings

Functionality Overview

KPI CALCULATION

A system task runs on the AMT server every hour to recalculate the statistics in the Dashboard. This process is postponed to the next hour if the load on the AMT server is too high.

SCHEDULED HOURS

If scheduled hours have not been set up for an equipment, the availability statistic in the Dashboard will report **NaN** (no data). Scheduled hours can be created using the function **Scheduled Hours**.

MULTIPLE UNITS OF MEASURE (UOMS)

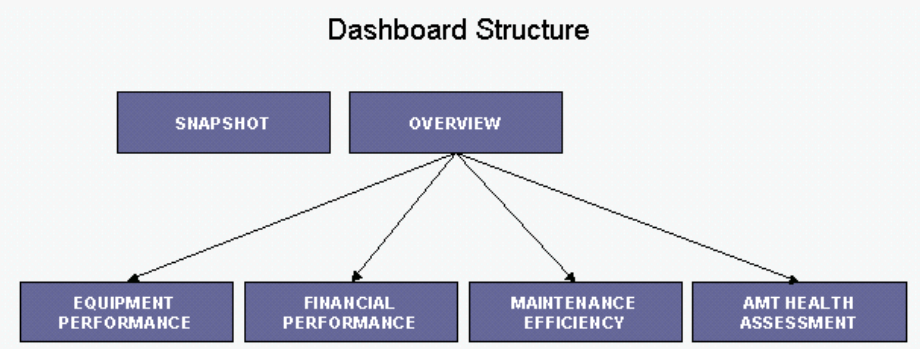
An equipment can be linked to a usage profile, which can contain multiple UOMs. If more than one of these UOMs is flagged in the **Units of Measure** system table as being **Operating Hours**, then this may cause usage-based statistics in the Dashboard to calculate inaccurately.

2. Hierarchy

Functionality Overview

AMT's Dashboard provides a real-time graphical view of performance indicators to help identify the areas that need attention. This includes both a short-term snapshot that focuses on the last 48 hours and a long-term overview of the month to date performance. Using Balance Scorecard principles, you can drill down both organisationally and functionally.

DASHBOARD STRUCTURE



The dashboard is not limited to equipment performance reporting and can be used for a wide variety of reporting. Being web-based, a range of users in your organisation can access it.

The actual display is like a spreadsheet, with colour indicating levels of performance. The colour scheme is:

Colour	Description
	Good (blue)
	Normal (white)
	Poor (yellow)
	Alert (red)
NaN	No Value (that is, no data on which to calculate this measurement)

At a glance, you can ascertain areas of poor performance that would require follow-up.

The following sections discuss elements of the dashboard in detail, beginning with navigation.

3. Navigation

Functionality Overview

Navigate within the dashboard using your mouse and standard internet browser buttons (such as Back, Forward, Print).

The opening screen, which is the snapshot, displays the total number of machines registered and a summary of key measurements. You can drill down into the measurements, from the highest level to the lowest (individual equipment plans).

Navigate to dashboard measurements is tabulated below.

Section	Group	Sub-Group	Measurement
Snapshot *Last 48 Hours*			Equipment Currently Down
			Availability
			Utilisation
			Strategy Tasks – Overdue
			Backlog Ratio
			WIP (days)
Overview	Equipment Performance		Utilisation
			Availability
			Reliability (MTBS)
			PCV%
	Financial Performance	Risk	Net Opportunity Risk
			% Of Projected Life Cycle Costs
		Financial Margin	Projected Billings
			Budget Margin %
			Projected Margin %
			Risk Adjustment Margin %
			Avg Margin %
		Cash Flow To Date	Cash Flow to Date
		AMT Health	AMT Health
	Maintenance Efficiency	Strategy	PCV%
			Strategy Tasks – Action Required

Section	Group	Sub-Group	Measurement
			Strategy Tasks – Overdue
		Planning	Backlog Ratio
			Planned DT % (DT = Downtime) #
			Service Accuracy#
		Daily Operations	Maintenance Ratio#
			Breakdown (Events) %#
			MTFS After PM#
			MTTR#
			WIP (days) @
		Customer Service	Budget Variance
	AMT Health		Budget Variance
			Strategy Tasks – Action Required
			Strategy Tasks – Overdue
			Strategy Tasks – Not linked to WOs

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

4. Calculations

Functionality Overview

The table below contains an alphabetical listing of all measurements reported in the dashboard, and a description of how the measurement is calculated, regardless of date range or time period.

The date range/time period for the measurement depends on the section of the dashboard in which the measurement is located. For example, measurements under Snapshot are calculated using data from the last 48 hours. Measurements under Overview are calculated from the month-to-date. The table below only describes the base calculation method, regardless of the time period.

Measurement	Calculation Method	OEM Specific
AMT Health	Average of the health indicators	
Availability	Depends on the availability set up in AMT for each fleet. PA = Physical availability, MA = Mechanical availability	
Backlog Ratio	Number of Backlog Tasks / Number of Equipment	
Breakdown % #	Number of Events where Breakdown = True / Total Number of Events	
Budget Margin %	(Projected Billing for the Budget Projection - Projected Cost@Cost for the Budget Projection) / Projected Billing for the Budget Projection	Y
Budget Variance	Projected Billing for the Budget Projection - Projected Cost@Sell not escalated for the Budget Projection	Y
Cash Flow to Date	Actual Billings to date – Work Orders Cost@Cost where Cost Bearer = < default Cost Bearer >	Y
Equipment Currently Down	(Number of Equipment Currently Down / Number of Equipment) * 100	
Maintenance Ratio #	Labour Hours / Equipment Usage	
MTFS After PM #	Mean Time to First Stoppage After PM. Total of Usage (in Engine Hrs) from PM Event to the Next Downtime Event/ Total Number of PM Events. A PM event is when there is a PM task in the event.	
MTTR #	Total Downtime for the Events / Total number of Events	
PCV%	(Projected Total Budget Projection - Projected Total Current Projection) / Projected Budget Projection	
Planned DT % **(DT = Downtime)** #	Total Downtime of events where Planned = True / Total Downtime of all events	
Projected Billings	Total projected billing based on the billings profile for the Current Projection	Y

Measurement	Calculation Method	OEM Specific
Projected Margin %	(Projected Billing for the Current Projection - Projected Cost@Cost for the Current Projection)/ Projected Billing for the Current Projection	Y
Reliability (MTBS)	Total Usage of the Equipment/ Total Number of Events	
Risk: Net Opportunity Risk	The sum of the Opportunity Components (that is, life used <20% at the end of the equipment life) – the sum of the Risk components (that is, life remaining <20% at the end of the equipment life). A positive value is greater opportunity than risk.	
Risk: % of Projected Life-Cycle Costs	Net Opportunity Risk / Projected Sell of Current Projection	
Risk: Adjustment Margin %	(Projected Billing for the Current Projection - Projected Cost@Cost for the Current Projection + Net Opportunity Risk Components) / Projected Billing for the Current Projection	Y
Risk: Avg Margin %	([Projected Margin %] + [Risk Adjust Margin %]) / 2	Y
Service Accuracy #	Number of services (PM Events) within the allowed Tolerance for a PM / Total Number of PM Events (Tolerance is set using Fleet Administration)	
Strategy Tasks – Action Required	Number of Strategy tasks that are either Overdue or (in the planning window and are system calculated and Not Reviewed) / Total Number of Strategy Tasks. **Note**: Only strategy tasks that are flagged as planning tasks are considered in this statistic.	
Strategy Tasks – Not linked to WOs	The number of strategy tasks in Completed or In Progress events in the last 12 months that are not linked to a work order.	
Strategy Tasks – Overdue	(Number of Strategy Tasks Overdue / Number of Strategy Tasks) * 100. Note: Only strategy tasks that are flagged as planning tasks are considered in this statistic.	
Utilisation	Usage / Number of Equipment	
WIP (days) @	(Total WIP Cost / Total Work Order Cost Last Year) * 365	Y

[#] *Rolling 3 month KPI.*

For measurements that are based on a date range greater than 1 month, the date range used includes prior months plus the month-to-date. For example, the last quarter is defined as the last 3 months as well as this month to-date. If the current date is 12-JUN-07, the last quarter is: 1-MAR-07 to 12-JUN-07 (March-April-May PLUS June to date).

Statistic is only relevant for original equipment manufacturers and their MARC contracts.

5. Audit Trail

Functionality Overview

Measurement	Source	Type
Availability	Availability (Mechanical) or Availability (Physical) • Responsibility = <default Responsibility>	Report
Backlog Ratio	Equipment Backlog Listing • Strategy Tasks = Yes	Report
Breakdown %	Breakdown Downtime Events % • Responsibility = <default Responsibility>	Report
Budget Margin %	GM Comparison	Report
Budget Variance	Equipment Plan Editor – Configuration – MARC	Screen
Cash Flow to Date	Cash Flow Statement	Report
Equipment Currently Down	EQS – Equipment Down	Screen
Maintenance Ratio	Maintenance Ratio • Cost Bearer = <default Cost Bearer>	Report
MTFS After PM	MTFS After PM • Responsibility = <default Responsibility> • PM Task Type = PM • FS Task Type = All • Planned = All • Breakdown = All	Report
MTTR	MTTR • Responsibility = <default Responsibility>	Report
PCV%	PCV Chart	Report
Planned DT %	Planned Downtime Hours % • Responsibility = <default Responsibility>	Report
Projected Billings	MARC PCV Chart	Report
Projected Margin %	GM Comparison	Report
Reliability (MTBS)	MTBS • Responsibility = <default Responsibility>	Report

Measurement	Source	Type
Net Opportunity Risk	High Risk High Opportunity • Default filters	Report
Risk: % of Projected Life-Cycle Costs	Dashboard calculation	
Risk: Adjustment Margin %	Dashboard calculation	
Risk: Avg Margin %	Dashboard calculation	
Service Accuracy	Service Accuracy	Report
Strategy Tasks – Action Required	Strategy Confirmation	Screen
Strategy Tasks – Not linked to WOs	EQS – Tasks – Admin Alerts	Screen
Strategy Tasks – Overdue	Overdue Strategy Tasks	Report
Utilisation	Usage Analysis Chart • Gross Usages As = Annual • Avg Per Eqp Plan = Yes	Report
WIP (days)	Work orders imported from the ERP system	Interface



For measurements that are based on a date range greater than 1 month, the date range used includes prior months plus the month-to-date. For example, the last quarter is defined as the last 3 months as well as this month to-date. If the current date is 12-JUN-24, the last quarter is: 1-MAR-24 to 12-JUN-24 (March-April-May PLUS June to date).

Learning Check



Check your understanding of the workshop and ask any related questions.

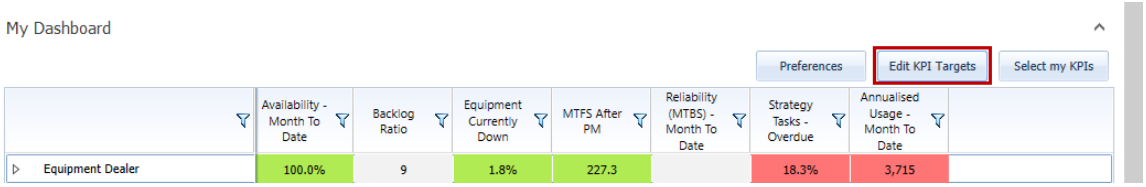
Notes

6. Targets

Procedure – Setting Default Targets

Steps to set targets at the fleet level:

1. Click **Edit KPI Targets**.



All measurements and corresponding default targets display.

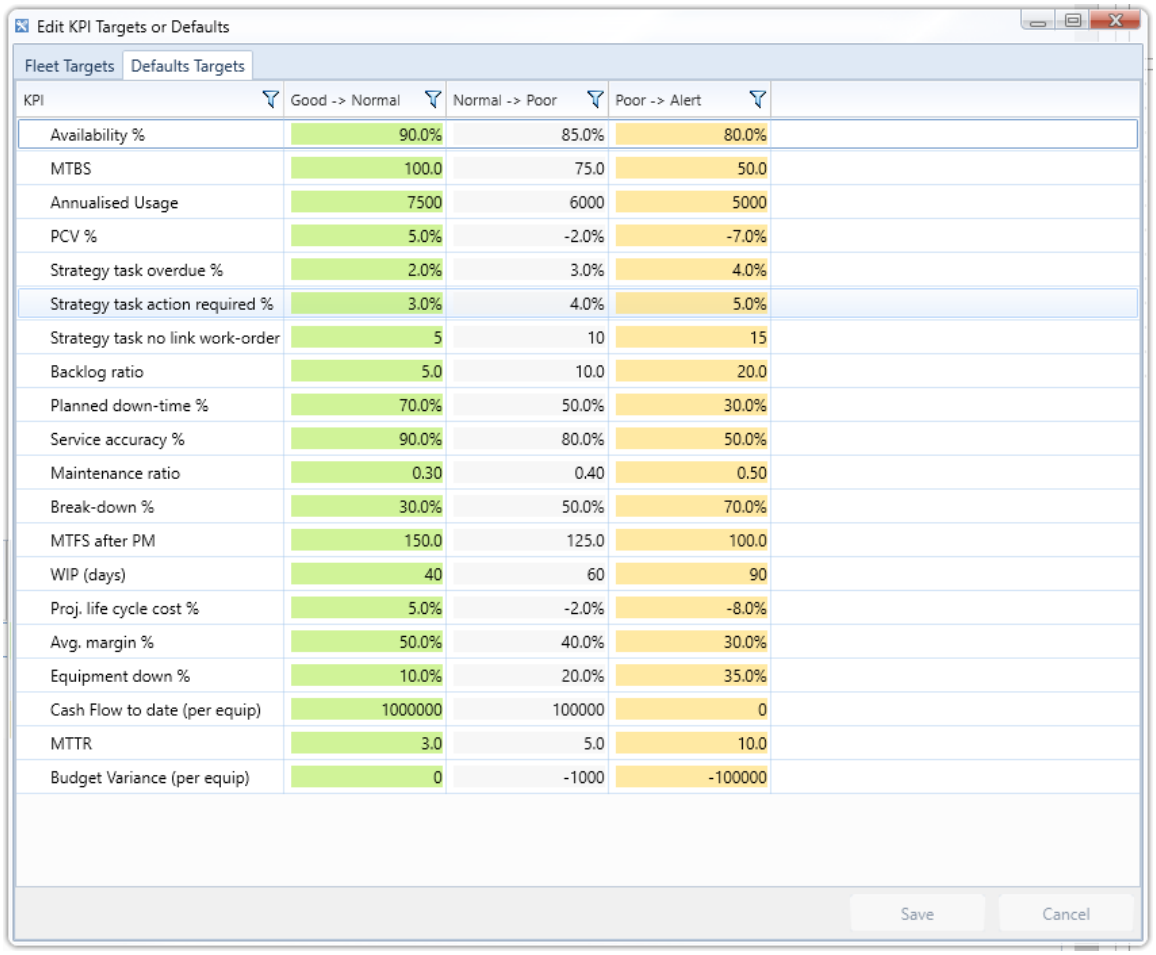
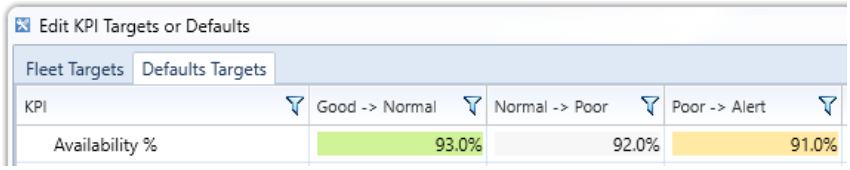


Figure 1. Edit KPI Targets or Defaults

2. Double-click on any **measurement** corresponding to the **KPI** whose targets you want to change.
3. Enter the new targets.

SET ACTUAL TARGETS EXAMPLE



In the above example, a rating of **Good** is given if the measurement is ≥ 93%. A rating of **Normal** is given if the measurement is < 93.0% and ≥ 92.0%. A rating of **Poor** is given if the measurement is < 92.0% and ≥ 91.0%. Finally, a rating of **Alert** is given if the measurement is < 91.0%.

Procedure – Setting Targets by Fleet

For every measurement, there are four targets (benchmarks) that define what range of calculation results equate to good, normal, poor, and alert performance. Desired targets can be applied to all equipment plans across all branches, or specific targets can be set at the fleet level.

1. Steps to set targets at the fleet level:
2. Click **Edit KPI Targets**.
3. In **Fleet Targets**, select a **Branch**, **Site** and **Fleet**.
4. All dashboard measurements and corresponding targets for the selected Branch, Site, and Fleet display.
5. Double-click on the measurement corresponding to the KPI that you want to change.
6. Enter the new targets.

Typically only the AMT Administrator has access to edit the targets.

Determining targets should be a joint decision between site, branch and head office management.

7. Printing

Functionality Overview

There are two methods of printing the dashboard:

USING YOUR INTERNET BROWSER

To print a hard copy of the dashboard, navigate within the dashboard to the measurements you want to print. Make sure the required measurements display on the screen and then, using your internet web browser, select **Print** from the browser's main menu or click the **Print** button on the browser's toolbar. What you see on your screen should be printed directly to the printer.

COPY/PASTE/CROP

Navigate to the measurements from the dashboard that you want to print. Take a screen shot by pressing the **Print Screen** key on your keyboard. This will save an image of what is currently showing on your screen and store it in the Windows Clipboard. Open your Word Processing application and click **Paste**. The image should now be visible on the page. Use a cropping tool to tighten the image as required.

Individual hardcopy reports for each of the measurements can be generated from within AMT. This gives you greater flexibility in specifying report types (for example, pie chart, text, line chart) and parameters such as date ranges, filters and analyse by criteria.



When printing the Dashboard using your internet web browser, you may have to change the Page Orientation/Set-up from Portrait to Landscape.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes